Asymptotic center–manifold for systems with parameter perturbations

Prabal S. Negi*
Okinawa Institute of Science and Technology Graduate University, Onna, Okinawa 904-0495, Japan
(Dated: September 4, 2025)

Asymptotic center-manifold approximations are constructed for systems that have been perturbed slightly away from the bifurcation point in the parameter space. This is done by extending the existing system to include parameter perturbations as dynamic variables, leading to an extended critical subspace and dimensionality of the center-manifold. The extension is done in a very general setting and it is shown that the consequences of such a system extension are non-trivial. The extended critical subspace has a certain structure associated with it, which may be deduced from the original system. The application of the center-manifold theorem to the extended system leads to a graph equation which rigorously valid in the vicinity of the bifurcation point (even in parameter space). The graph equation is approximated as an asymptotic power series in the amplitudes of the critical eigenmodes leading to general expressions for restricted resolvent solutions for the various terms in the power series at each order. A Ginzburg-Landau model near bifurcation is used to demonstrate the validity of the asymptotic center-manifold in the extended space.

I. PROBLEM DEFINITION

A. The Standard System

Consider a system with its state variable denoted by $\boldsymbol{U}$, consisting of l parameters given by the vector $\boldsymbol{\mu}$, and an evolution law given by

$$\frac{\partial \boldsymbol{U}}{\partial t} = \boldsymbol{f}(\boldsymbol{U}, \boldsymbol{\mu}). \tag{1}$$

The evolution law may be a finite set of ordinary differential equations or, an infinite dimensional partial differential equation as often arises for the space-time evolution of physical systems. Typically, physical systems will have real parameters $\boldsymbol{\mu}$ however, allowing for complex parameters presents no additional difficulty and we assume that $\boldsymbol{\mu} \in \mathbb{C}^l$. For a certain set of parameter values $\boldsymbol{\mu} = \boldsymbol{\mu^b}$, the system is assumed to have a fixed point for a state vector $\boldsymbol{U^b}$ satisfying

$$0 = \boldsymbol{f}(\boldsymbol{U^b}, \boldsymbol{\mu^b}). \tag{2}$$

The system may be rewritten in terms of the deviation from the fixed point solution $\boldsymbol{U} = \boldsymbol{U^b} + \boldsymbol{u}$ such that

$$\begin{aligned}\frac{\partial \boldsymbol{u}}{\partial t} =& \mathbf{L}\boldsymbol{u} + \\ & [\boldsymbol{f}(\boldsymbol{U^b} + \boldsymbol{u}, \boldsymbol{\mu^b}) - \mathbf{L}\boldsymbol{u}],\end{aligned} \tag{3}$$

where, $\mathbf{L}$ is the linearized operator at the fixed point, acting on $\boldsymbol{u}$, defined by

$$\mathbf{L}\boldsymbol{u} = \left(\frac{\partial \boldsymbol{f}}{\partial \boldsymbol{U}}\bigg|_{(\boldsymbol{U^b}, \boldsymbol{\mu^b})} \right) \cdot \boldsymbol{u}. \tag{4}$$

The right hand side in equation (3) has been divided into the linear and non-linear contributions with $\mathbf{L}(\boldsymbol{u})$ representing the linear part and the remaining terms in the square brackets representing the non-linear contributions.

Furthermore, we assume that $\boldsymbol{\mu^b}$ represents the parameters of the system at a bifurcation point. It is generically assumed that the system undergoes a multiple codimension bufurcation at $\boldsymbol{\mu^b}$ and that the linearized operator $\mathbf{L}$ has a discrete set of n critical eigenvalues λ^C, with zero growth rate ($\mathfrak{Re}(\lambda^C) = 0$). This includes both purely real critical eigenvalues with $\mathfrak{Re}(\lambda^C) = 0$, $\mathfrak{Im}(\lambda^C) = 0$, and purely imaginary critical eigenvalues with $\mathfrak{Re}(\lambda^C) = 0$, $\mathfrak{Im}(\lambda^C) \neq 0$. For real systems complex eigenvalues occur in conjugate pairs. Accordingly, the system has n critical eigenpairs $(\lambda_i^C, \boldsymbol{v}_i^C), i \in 1 \ldots n$, where $\boldsymbol{v}_i^C$ represents the eigenvector corresponding to λ_i^C.

The adjoint of the direct linearized operator $\mathbf{L}$ is denoted as $\mathbf{L}^\dagger$ which also has n critical eigenpairs $(\lambda_i^{C*}, \boldsymbol{w}_i^C)$. The scaling of the eigenvectors of the direct and adjoint operators may be defined such that they satisfy the birthogonality condition

$$\langle \boldsymbol{w}_i^C, \boldsymbol{v}_j^C \rangle = \boldsymbol{w}_i^{C\dagger} \boldsymbol{v}_j^C = \delta_{i,j}, \tag{5}$$

where $\langle \cdot, \cdot \rangle$ is the standard inner-product and $\delta_{i,j}$ is the Kronecker-delta fuction. The superscript † is used here to indicate a complex-conjugated transpose of vectors and matrix operators. For scalars complex-conjugation is indicated by a * superscript. The superscript ^{H} which is often used for the same purpose is reserved for objects referring to harmonic forcings. One may define a matrix $\mathbf{V}$, whose columns comprise of the n critical eigenvectors of $\mathbf{L}$ and the matrix spans the invariant center subspace of the linear operator $\mathbf{L}$. The corresponding matrix for the adjoint eigenvectors is denoted $\mathbf{W}$. Due to the chosen scaling denoted by equation (5) the matrices satisfy,

$$\mathbf{W}^\dagger \mathbf{V} = \mathbf{I}_n, \tag{6}$$

where, $\mathbf{I}_n$ represents an identity matrix of order $n \times n$. The projection operator into the critical subspace (also the center subspace) is defined by $\mathbf{P} = \mathbf{V}\mathbf{W}^\dagger$, while the

* prabal.negi@oist.jp

projection operator into its complement (stable subspace of $\mathbf{L}$) is defined by, $\mathbf{Q} = \mathbf{I} - \mathbf{V}\mathbf{W}^\dagger$.

Additionally, $\mathbf{\Lambda}_C$ is defined as an $n \times n$ diagonal matrix consisting of the critical eigenvalues of $\mathbf{L}$, with the implicit assumption of diagonalizability of $\mathbf{L}$ within this subspace. The ordering of the eigenvalues in $\mathbf{\Lambda}_C$ and eigenvectors in $\mathbf{V}$ is consistent with each other such that $\mathbf{L}\mathbf{V} = \mathbf{V}\mathbf{\Lambda}_C$.

B. The Extended System

The same system is now considered for the case of parameter perturbation so that the evolution dynamics are now evaluated at parameter values of $\boldsymbol{\mu} = \boldsymbol{\mu^b} + \boldsymbol{\nu}$, with $\boldsymbol{\nu}$ being the vector of parameter perturbations, $\boldsymbol{\nu} = [\nu_1, \nu_2, \ldots, \nu_l]$. In addition, the system is forced with small-amplitude harmonic forcing of the type

$$\boldsymbol{f}_H(t) = \sum_{i=1}^{h} \boldsymbol{f}_i \theta_i(t); \qquad \theta_i = \epsilon_i e^{\lambda_i^H t}, \tag{7}$$

where, the various θ_i are treated as the dynamic variable, λ_i^H is purely imaginary, with the imaginary part representing the angular frequency of the forcing. $\boldsymbol{f}_i$ represents the shape of the i^{th} harmonic forcing and ϵ_i is a scale factor. Constant forcing terms are readily incorporated when a particular (or multiple) $\lambda_i^H = 0$. The vector of the various θ_i is denoted as $\boldsymbol{\theta} = [\theta_1; \theta_2; \ldots; \theta_h]$, and the vector of angular frequencies is denoted as $\boldsymbol{\lambda}^H = [\lambda_1^H; \lambda_2^H; \ldots; \lambda_h^H]$. The resulting evolution equation is

$$\frac{\partial \boldsymbol{u}}{\partial t} = \boldsymbol{f}(\boldsymbol{U^b} + \boldsymbol{u}, \boldsymbol{\mu^b} + \boldsymbol{\nu}) + \sum_{i=1}^{h} \boldsymbol{f}_i \theta_i; .$$

The diagonal matrix for the harmonic angular frequencies is represented by $\mathbf{\Lambda}_H = diagm(\boldsymbol{\lambda}^H)$. Similarly, $\boldsymbol{\lambda}^P = [\lambda_1^P; \lambda_2^P; \ldots; \lambda_l^P]$ is defined as a vector of length l, with all $\lambda_i^P = 0$ identically, for $i \in 1 \ldots l$. These may be thought of parameter eigenvalues of parameter angular frequencies depending on the point of view. In either case, these vanish identically. $\mathbf{\Lambda}_P = diagm(\boldsymbol{\lambda}^P)$ is defined as an $l \times l$ null matrix, to represent the trivial linear evolution matrix for the parameter variables $\boldsymbol{\nu}$. The extended system including the evolution equations for the newly introduced variables $\boldsymbol{\nu}$ and $\boldsymbol{\theta}$, henceforth referred to as the *extended variables*, is given by,

$$\frac{\partial \boldsymbol{u}}{\partial t} = \mathbf{L}\boldsymbol{u} + \mathbf{L}_{\boldsymbol{\mu}}\boldsymbol{\nu} + \mathbf{L}_{\boldsymbol{\theta}}\boldsymbol{\theta} + \boldsymbol{f}_{NL}(\boldsymbol{u}, \boldsymbol{\nu}) \tag{8a}$$

$$\frac{d\boldsymbol{\nu}}{dt} = \mathbf{\Lambda}_P \boldsymbol{\nu} \tag{8b}$$

$$\frac{d\boldsymbol{\theta}}{dt} = \mathbf{\Lambda}_H \boldsymbol{\theta}, \tag{8c}$$

Here, as done earlier, the evolution equations are split into the linear and non-linear contributions. $\mathbf{L}$ is the same linearized operator as defined earlier in equation 4. $\mathbf{L}_{\boldsymbol{\mu}}$ is the linearized operator acting on the newly introduced extended variable $\boldsymbol{\nu}$, defined as

$$\mathbf{L}_{\boldsymbol{\mu}}\boldsymbol{\nu} = \left(\left. \frac{\partial \boldsymbol{f}}{\partial \boldsymbol{\mu}} \right|_{(\boldsymbol{U^b}, \boldsymbol{\mu^b})} \right) \cdot \boldsymbol{\nu}. \tag{9}$$

Similarly, $\mathbf{L}_{\boldsymbol{\theta}}$ is the linearized operator acting on $\boldsymbol{\theta}$, defined as,

$$\mathbf{L}_{\boldsymbol{\theta}}\boldsymbol{\theta} = \left(\frac{\partial \boldsymbol{f}_H}{\partial \boldsymbol{\theta}} \right) \cdot \boldsymbol{\theta}. \tag{10}$$

This is the same expression as in equation (7), just written in a manner consistent with the other terms. Finally, $\boldsymbol{f}_{NL}(\boldsymbol{u}, \boldsymbol{\nu})$ represents the non-linear terms of the modified system given by

$$\begin{aligned} \boldsymbol{f}_{NL}(\boldsymbol{u}, \boldsymbol{\nu}) =& \boldsymbol{f}(\boldsymbol{U^b} + \boldsymbol{u}, \boldsymbol{\mu^b} + \boldsymbol{\nu}) \\ & - \mathbf{L}\boldsymbol{u} - \mathbf{L}_{\boldsymbol{\mu}}\boldsymbol{\nu}, \end{aligned} \tag{11}$$

where, the dependence on $\boldsymbol{U^b}$ and $\boldsymbol{\mu^b}$ has been suppressed for notational convenience.

An overhead hat notation, $\widehat{\cdot}$, is used to refer to variables and operators associated with the extend space, so that the new state vector, non-linear term and the linearized direct and adjoint operators are denoted by $\widehat{\boldsymbol{u}}$, $\widehat{\boldsymbol{f}}_{NL}$, $\widehat{\mathcal{L}}$ and $\widehat{\mathcal{L}^\dagger}$, that are defined as,

$$\widehat{\boldsymbol{u}} = \left[\boldsymbol{u};\ \boldsymbol{\nu};\ \boldsymbol{\theta}\right],$$
$$\widehat{\boldsymbol{f}}_{NL} = \left[\boldsymbol{f}_{NL};\ \mathbf{0};\ \mathbf{0}\right],$$

$$\widehat{\mathbf{L}} = \begin{bmatrix} \mathbf{L} & \mathbf{L}_{\boldsymbol{\mu}} & \mathbf{L}_{\boldsymbol{\theta}} \\ \mathbf{0} & \mathbf{\Lambda}_P & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{\Lambda}_H \end{bmatrix}, \quad \widehat{\mathbf{L}}^\dagger = \begin{bmatrix} \mathbf{L}^\dagger & \mathbf{0} & \mathbf{0} \\ \mathbf{L}_{\boldsymbol{\mu}}^\dagger & \mathbf{\Lambda}_P^\dagger & \mathbf{0} \\ \mathbf{L}_{\boldsymbol{\theta}}^\dagger & \mathbf{0} & \mathbf{\Lambda}_H^\dagger \end{bmatrix}, \tag{12}$$

where, the superscript ^{T} refers to vector transpose. The full system may then be represented compactly as

$$\frac{\partial \widehat{\boldsymbol{u}}}{\partial t} = \widehat{\mathbf{L}}\widehat{\boldsymbol{u}} + \widehat{\boldsymbol{f}}_{NL}. \tag{13}$$

The system defined by equation (8), or its compact form equation (13), with additional dynamical equations for the extended variables will be referred to as the *extended system.*

C. Extended Subspace Structure

The extended system has a larger critical subspace of dimension $m = n + l + h$, as compared to the original system due to the $l + h$ additional variables introduced. While the extension of the system seems almost trivial, the eigenvector structure in the additional critical dimensions however is far from trivial. Nevertheless, the extended eigenspace structure can be deduced directly from the properties of the original system. It turns out to be incredibly useful to analyze the structure of the extended subspace since the the adjoint extended critical subspace does have a special structure which allows the asymptotic terms of the center-manifold to be evaluated using the original linear system $\mathbf{L}$.

There are however some subtleties involved when the system extension introduces new eigenvalues that are resonant with the original linear system $\mathbf{L}$. In such a scenario the system extension leads to a generalized eigenspace for the resonant eigenvalues. The subspace structure is now evaluated in detail for both the resonant and non-resonant cases. This is first done for the extended operator $\widehat{\mathbf{L}}$ and then for the adjoint operator $\widehat{\mathbf{L}}^\dagger$.

1. Subspace structure for $\widehat{\mathbf{L}}$

Observe that if $(\lambda_i^C, \boldsymbol{v}_i^C)$ is a critical eigenpair of the original linearized operator $\mathbf{L}$, then its extension to $\widehat{\mathbf{L}}$ satisfies,

$$(\lambda_i^C \mathbf{I} - \widehat{\mathbf{L}})\widehat{\boldsymbol{v}}_i^C = \mathbf{0}, \tag{14a}$$
$$\widehat{\boldsymbol{v}}_i^C = \left[\boldsymbol{v}_i^C;\ \mathbf{0};\ \mathbf{0}\right], \tag{14b}$$

$\forall i \in 1, \ldots, n$, which may be verified via simple substitution. Hence all critical eigenmodes of $\mathbf{L}$ can be trivially extended to $\widehat{\mathbf{L}}$. This definition for the extension of the eigenvectors in fact applies even to modes that are not critical eigenmodes.

The additional critical eigenpairs introduced due to system extension need to be deduced. Two sets of canonical unit vectors are defined such that,

$$\boldsymbol{e}_i^P = [\overbrace{0; 0; \ldots; 0}^{i-1}; 1; \overbrace{0; 0; \ldots; 0}^{l-i}], \tag{15a}$$
$$\boldsymbol{e}_i^H = [\underbrace{0; 0; \ldots; 0}_{i-1}; 1; \underbrace{0; 0; \ldots; 0}_{h-i}]. \tag{15b}$$

If the original linear system $\mathbf{L}$ is non-singular then, the l eigenpairs generated by the system extension due to $\boldsymbol{\nu}$ have the structure of the type $(\lambda_i^P, [\boldsymbol{v}_i^P; \boldsymbol{e}_i^P; \mathbf{0}])$, which satisfy

$$(\lambda_i^P \mathbf{I} - \mathbf{L})\boldsymbol{v}_i^P = \mathbf{L}_{\boldsymbol{\mu}}(\boldsymbol{e}_i^P), \qquad (\lambda_i^P = 0) \tag{16a}$$
$$\widehat{\boldsymbol{v}}_i^P = \left[\boldsymbol{v}_i^p;\ \boldsymbol{e}_i^P;\ \mathbf{0}\right], \quad \forall i \in 1, \ldots l. \tag{16b}$$

This can be verified that $(\lambda_i^P - \widehat{\mathbf{L}})\widehat{\boldsymbol{v}}_i^P = 0$.

However, if $\mathbf{L}$ is singular then equation (16) is indeterminate and the definition needs to be modified. For the i^{th} *parameter mode*, define a vector $\boldsymbol{\gamma}_P^i \in \mathbb{C}^n$, where the j^{th} element of the vector is given by,

$$(\boldsymbol{\gamma}_i^P)_j = \begin{cases} \boldsymbol{w}_j^{C\dagger}\mathbf{L}_{\boldsymbol{\mu}}(\boldsymbol{e}_i^P), & \textit{if } \lambda_i^P = \lambda_j^C \\ 0, & \textit{otherwise.} \end{cases} \tag{17}$$

$$(\lambda_i^P \mathbf{I} - \mathbf{L})\boldsymbol{v}_i^P = \mathbf{L}_{\boldsymbol{\mu}}(\boldsymbol{e}_i^P) - \mathbf{V}\boldsymbol{\gamma}_i^P, \quad (\lambda_i^P = 0) \tag{18a}$$
$$\boldsymbol{w}_j^{C\dagger}\boldsymbol{v}_i^P = 0, \qquad \textit{if } \lambda_i^P = \lambda_j^C, \tag{18b}$$
$$\widehat{\boldsymbol{v}}_i^P = \left[\boldsymbol{v}_i^P;\ \boldsymbol{e}_i^P;\ \mathbf{0}\right], \quad \forall i \in 1, \ldots l. \tag{18c}$$

In the singular case, $\widehat{\boldsymbol{v}}_i^P$ is no longer an eigenvector of $\widehat{\mathbf{L}}$, however, it is a generalized eigenvector satisfying $(\lambda_i^P \mathbf{I} - \widehat{\mathbf{L}})^2 \widehat{\boldsymbol{v}}_i^P = \mathbf{0}$. Note that $\boldsymbol{\gamma}_i^P$ is identically zero for the non-singular case and equation (18) reduces to equation (16). Henceforth these modes shall be referred to as *parameter modes* and will be defined by equation (18), since it is the more general definition. The matrix of all the vectors $\boldsymbol{\gamma}_i^P$ is denoted $\boldsymbol{\Gamma}_P$.

Similarly, when the harmonic forcing is non-resonant, the h eigenpairs generated by system extension due to $\boldsymbol{\theta}$ have the following structure,

$$(\lambda_i^H \mathbf{I} - \mathbf{L})\boldsymbol{v}_i^H = \mathbf{L}_{\boldsymbol{\theta}}(\boldsymbol{e}_i^H), \tag{19a}$$
$$\widehat{\boldsymbol{v}}_i^H = \left[\boldsymbol{v}_i^H;\ \mathbf{0};\ \boldsymbol{e}_i^H\right], \quad \forall i \in 1, \ldots h. \tag{19b}$$

On the other hand, when the harmonic forcing is resonant with some of the angular frequencies of $\mathbf{L}$ then, analogous to the case of parameter modes, one defines,

$$(\boldsymbol{\gamma}_i^H)_j = \begin{cases} \boldsymbol{w}_j^{C\dagger}\mathbf{L}_{\boldsymbol{\theta}}(\boldsymbol{e}_i^H), & \textit{if } \lambda_i^H = \lambda_j^C \\ 0, & \textit{otherwise,} \end{cases} \tag{20}$$

where, $(\boldsymbol{\gamma}_i^H)_j$ is the j^{th} component of the vector $\boldsymbol{\gamma}_i^H$. The extended modes are then given by

$$(\lambda_i^H \mathbf{I} - \mathbf{L})\boldsymbol{v}_i^H = \mathbf{L}_{\boldsymbol{\theta}}(\boldsymbol{e}_i^H) - \mathbf{V}\boldsymbol{\gamma}_i^H, \tag{21a}$$
$$\boldsymbol{w}_j^{C\dagger}\boldsymbol{v}_i^H = 0, \qquad \textit{if } \lambda_i^H = \lambda_j^C, \tag{21b}$$
$$\widehat{\boldsymbol{v}}_i^H = \left[\boldsymbol{v}_i^H;\ \mathbf{0};\ \boldsymbol{e}_i^H\right], \quad \forall i \in 1, \ldots h. \tag{21c}$$

In the resonant case, this satisfies the generalized eigenvalue equation $(\lambda_i^H \mathbf{I} - \widehat{\mathbf{L}})^2 \widehat{\boldsymbol{v}}_i^H = \mathbf{0}$. Again, equation (21) reduces to equation (19) when the forcing is non-resonant. Henceforth these modes will be referred to as *forcing modes* and are defined by the more general definition given by equation (21). The matrix of all the vectors $\boldsymbol{\gamma}_i^H$ is denoted $\boldsymbol{\Gamma}_H$. The critical eigenspace of $\widehat{\mathbf{L}}$ can be represented by three matrices

$$\widehat{\mathbf{V}}_C = \begin{bmatrix} \mathbf{V} \\ \mathbf{0} \\ \mathbf{0} \end{bmatrix}, \quad \widehat{\mathbf{V}}_P = \begin{bmatrix} \mathbf{V}_P \\ \mathbf{I} \\ \mathbf{0} \end{bmatrix}, \quad \widehat{\mathbf{V}}_H = \begin{bmatrix} \mathbf{V}_H \\ \mathbf{0} \\ \mathbf{I} \end{bmatrix}, \tag{22}$$

where, $\mathbf{V}$ is the matrix of the critical eigenvectors of $\mathbf{L}$. The matrix $\mathbf{V}_P$ comprises of the l vectors $\boldsymbol{v}_i^P$ of the parameter modes, and, $\mathbf{V}_H$ comprises of the h vectors $\boldsymbol{v}_i^H$ of the forcing modes. The matrix of vectors of the extended eigenspace is denoted by, $\widehat{\mathbf{V}} = [\widehat{\mathbf{V}}_C, \widehat{\mathbf{V}}_P, \widehat{\mathbf{V}}_H]$.

2. Subspace structure for $\widehat{\mathbf{L}}^\dagger$

The eigenspace structure for the adjoint problem is important to consider since these vectors define the projection operators into the extended critical subspace. The critical eigenmodes of the $\widehat{\mathbf{L}}^\dagger$ can be extended via the following definition,

$$(\lambda_i^{C*}\mathbf{I}-\mathbf{L}^\dagger)\boldsymbol{w}_i^C=0, \qquad \forall i\in 1,\dots n, \tag{23a}$$
$$\widehat{\boldsymbol{w}}_i^C=\left[\boldsymbol{w}_i^C;\ \boldsymbol{\zeta}_i^P;\ \boldsymbol{\zeta}_i^H\right], \tag{23b}$$

with vectors $\boldsymbol{\zeta}_i^P$ and $\boldsymbol{\zeta}_i^H$ defined such that the j^{th} component of $\boldsymbol{\zeta}_i^P$ given by,

$$(\boldsymbol{\zeta}_i^P)_j=\begin{cases}(\mathbf{L}_{\boldsymbol{\mu}}^\dagger\boldsymbol{w}_i^C)_j/(\lambda_i^{C*}-\lambda_j^{P*}); & \textit{if } \lambda_i^{C*}\neq\lambda_j^{P*},\\ 0; & \textit{if } \lambda_i^{C*}=\lambda_j^{P*},\end{cases} \tag{24}$$

$\forall j\in 1,\dots,l$, and, the j^{th} component of $\boldsymbol{\zeta}_i^H$ is given by,

$$(\boldsymbol{\zeta}_i^H)_j=\begin{cases}(\mathbf{L}_{\boldsymbol{\theta}}^\dagger\boldsymbol{w}_i^C)_j/(\lambda_i^{C*}-\lambda_j^{H*}); & \textit{if } \lambda_i^{C*}\neq\lambda_j^{H*},\\ 0; & \textit{if } \lambda_i^{C*}=\lambda_j^{H*},\end{cases} \tag{25}$$

$\forall j\in 1,\dots,h$. Here, $(\mathbf{L}_{\boldsymbol{\mu}}^\dagger\boldsymbol{w}_i^C)_j$ refers to the j^{th} component of the vector $(\mathbf{L}_{\boldsymbol{\mu}}^\dagger\boldsymbol{w}_i^C)$, and likewise for $(\mathbf{L}_{\boldsymbol{\theta}}^\dagger\boldsymbol{w}_i^C)_j$. It may be verified that the definition is consistent with the eigenvalue equation, $(\lambda_i^{C*}\mathbf{I}-\widehat{\mathbf{L}}^\dagger)\widehat{\boldsymbol{w}}_i^C=0$, for the non-resonant cases, and satisfies generalized eigenvalue equation, $(\lambda_i^{C*}\mathbf{I}-\widehat{\mathbf{L}}^\dagger)^2\widehat{\boldsymbol{w}}_i^C=0$, for the resonant cases.

The adjoint parameter modes are given simply by,

$$\widehat{\boldsymbol{w}}_i^P=\left[\mathbf{0};\ \boldsymbol{e}_i^P;\ \mathbf{0}\right], \qquad \forall i\in 1,\dots l, \tag{26}$$

which satisfies $(\lambda_i^{P*}\mathbf{I}-\widehat{\mathbf{L}}^\dagger)\widehat{\boldsymbol{w}}_i^P=0$. And the adjoint forcing modes are given by,

$$\widehat{\boldsymbol{w}}_i^H=\left[\mathbf{0};\ \mathbf{0};\ \boldsymbol{e}_i^H\right], \qquad \forall i\in 1,\dots h, \tag{27}$$

which satisfies $(\lambda_i^{H*}\mathbf{I}-\widehat{\mathbf{L}}^\dagger)\widehat{\boldsymbol{w}}_i^H=0$.

The extended eigenspace of $\widehat{\mathbf{L}}^\dagger$ can be represented with matrices as

$$\widehat{\mathbf{W}}_C=\begin{bmatrix}\mathbf{W}\\ \mathbf{Z}_P\\ \mathbf{Z}_H\end{bmatrix}, \quad \widehat{\mathbf{W}}_P=\begin{bmatrix}\mathbf{0}\\ \mathbf{I}\\ \mathbf{0}\end{bmatrix}, \quad \widehat{\mathbf{W}}_H=\begin{bmatrix}\mathbf{0}\\ \mathbf{0}\\ \mathbf{I}\end{bmatrix}, \tag{28}$$

where, $\mathbf{W}$ is the matrix of the adjoint critical eigenvectors of $\mathbf{L}^\dagger$. The matrix $\mathbf{Z}_P$ comprises of the l vectors $\boldsymbol{\zeta}_i^P$ and, $\mathbf{Z}_H$ comprises of the h vectors $\boldsymbol{\zeta}_i^H$. The matrix of vectors of the critical adjoint eigenspace is denoted by, $\widehat{\mathbf{W}}=[\widehat{\mathbf{W}}_C,\widehat{\mathbf{W}}_P,\widehat{\mathbf{W}}_H]$. It can be verified that,

$$\widehat{\mathbf{W}}^\dagger\widehat{\mathbf{V}}=\mathbf{I}_m, \tag{29}$$

where, $m=n+l+h$. This is the analogue of equation (6), and is simply a statement of the biorthogonality relation between the direct and adjoint eigenvectors in the extended space. The projection matrices into the critical and stable subspaces of $\widehat{\mathbf{L}}$ are given by,

$$\widehat{\mathbf{P}}=\widehat{\mathbf{V}}\widehat{\mathbf{W}}^\dagger \tag{30}$$
$$\implies \widehat{\mathbf{P}}=\widehat{\mathbf{V}}_{\mathbf{C}}\widehat{\mathbf{W}}_C^\dagger+\widehat{\mathbf{V}}_{\mathbf{P}}\widehat{\mathbf{W}}_P^\dagger+\widehat{\mathbf{V}}_{\mathbf{H}}\widehat{\mathbf{W}}_H^\dagger,$$
$$\widehat{\mathbf{Q}}=\mathbf{I}-\widehat{\mathbf{V}}\widehat{\mathbf{W}}^\dagger, \tag{31}$$
$$\implies \widehat{\mathbf{Q}}=\mathbf{I}-\widehat{\mathbf{V}}_{\mathbf{C}}\widehat{\mathbf{W}}_C^\dagger-\widehat{\mathbf{V}}_{\mathbf{P}}\widehat{\mathbf{W}}_P^\dagger-\widehat{\mathbf{V}}_{\mathbf{H}}\widehat{\mathbf{W}}_H^\dagger.$$

Finally, a reduced matrix representing the oblique projection of the extended linear operator $\widehat{\mathbf{L}}$ onto the critical subspace is defined by,

$$\widehat{\mathbf{K}}=\widehat{\mathbf{W}}^\dagger\widehat{\mathbf{L}}\widehat{\mathbf{V}}=\begin{bmatrix}\boldsymbol{\Lambda}_C & \boldsymbol{\Gamma}_P & \boldsymbol{\Gamma}_H\\ \mathbf{0} & \boldsymbol{\Lambda}_P & \mathbf{0}\\ \mathbf{0} & \mathbf{0} & \boldsymbol{\Lambda}_H\end{bmatrix}, \tag{32}$$

which is a complex matrix of size $m\times m$. The diagonal elements of $\widehat{\mathbf{K}}$ represent the eigenvalues of the reduced system, that are the critical eigenvalues of the extended system, and may be represented by the vector $\widehat{\boldsymbol{\lambda}}=diag(\widehat{\mathbf{K}})$.

Note that due to the special structure of the adjoint parameter and forcing modes, the projection of the extended nonlinearity $\widehat{\boldsymbol{f}}_{NL}$ onto the subspace spanned by the (direct) parameter and forcing modes vanishes identically. That is to say $\widehat{\mathbf{W}}_P^\dagger\widehat{\boldsymbol{f}}_{NL}=\mathbf{0}$, and $\widehat{\mathbf{W}}_H^\dagger\widehat{\boldsymbol{f}}_{NL}=\mathbf{0}$.

This completes the extension of the system and the subspace structure for both the direct and adjoint systems can be readily identified. While the structure of the extended subspaces has been evaluated, reformulation of the problem as an extended system may have detrimental practical implications when considering large systems. Practical methods for solving such systems often rely on numerical algorithms that exploit inherent structure of the problem (Hermicity for conjugate-gradient solvers for example). In such a scenario, additional degrees of freedom can potential destroy the system symmetries and may require substantial changes in the numerical approach to the system solution. This would be particularly true of asymptotic approaches that require resolvent calculations [? ? ? ? ? ? ? ? ?], which itself is a computationally challenging task for large systems. Such a scenario would be highly undesireable. However, as it turns out, at least some of the parameterizations involved in the evaluation of center-manifolds do not in fact require the use of the extended system after the initial evaluation of the extended tangent space. All higher order calculations can be performed using the original linear system $\mathbf{L}$. This is clarified in the next section where the parameterization and its asymptotic terms are explicitly built. First however, we prove the following results which will come in handy.

Lemma 1. *Given an extended system described by equation* (8)*, with a linearized operator* $\widehat{\mathbf{L}}$*, given by equation* (12)*, if a vector,* $\widehat{\boldsymbol{u}}=[\boldsymbol{u};\boldsymbol{\nu};\boldsymbol{\theta}]$*, is such that its projection onto the center-subspace of* $\widehat{\mathbf{L}}$ *vanishes, then, the extended variables* $\boldsymbol{\nu}$ *and* $\boldsymbol{\theta}$ *must also vanish identically for* $\widehat{\boldsymbol{u}}$*.*

Proof. A vector $\widehat{\boldsymbol{u}}$ whose components along the center-subspace vanish implies that $\widehat{\mathbf{W}}^\dagger\widehat{\boldsymbol{u}}=\mathbf{0}$, where $\widehat{\mathbf{W}}=[\widehat{\mathbf{W}}_C,\widehat{\mathbf{W}}_P,\widehat{\mathbf{W}}_H]$, are as deduced in sections I B and I C.

$$\therefore \widehat{\mathbf{W}}^\dagger\widehat{\boldsymbol{u}}=\begin{bmatrix}\mathbf{W}^\dagger\boldsymbol{u}+\mathbf{Z}_P^\dagger\boldsymbol{\nu}+\mathbf{Z}_H^\dagger\boldsymbol{\theta}\\ \mathbf{I}^\dagger\boldsymbol{\nu}\\ \mathbf{I}^\dagger\boldsymbol{\theta}\end{bmatrix}=\begin{bmatrix}\mathbf{0}\\ \mathbf{0}\\ \mathbf{0}\end{bmatrix}.$$

For this to hold,

$$\mathbf{I}^{\dagger}\boldsymbol{\nu} = \mathbf{0}, \implies \boldsymbol{\nu} = \mathbf{0},$$
$$\mathbf{I}^{\dagger}\boldsymbol{\theta} = \mathbf{0}, \implies \boldsymbol{\theta} = \mathbf{0}.$$

Hence, both the extended variables $\boldsymbol{\nu}$ and $\boldsymbol{\theta}$ vanish identically. □

Corollary 1. *Given an extended system described by equation* (8)*, with a linearized operator* $\widehat{\mathbf{L}}$*, given by equation* (12)*, if a vector,* $\widehat{\boldsymbol{u}} = [\boldsymbol{u}; \boldsymbol{\nu}; \boldsymbol{\theta}]$*, is such that its projection onto the center-subspace of* $\widehat{\mathbf{L}}$ *vanishes, then, the projection of* $\boldsymbol{u}$ *onto the center-subspace of* $\mathbf{L}$ *also vanishes.*

Proof. For a vector $\widehat{\boldsymbol{u}}$ that satisfies $\widehat{\mathbf{W}}^{\dagger}\widehat{\boldsymbol{u}} = \mathbf{0}$, by lemma 1, both $\boldsymbol{\nu}$ and $\boldsymbol{\theta}$ vanish identically.

$$\therefore \widehat{\mathbf{W}}^{\dagger}\widehat{\boldsymbol{u}} = \begin{bmatrix} \mathbf{W}^{\dagger}\boldsymbol{u} \\ \mathbf{0} \\ \mathbf{0} \end{bmatrix} = \begin{bmatrix} \mathbf{0} \\ \mathbf{0} \\ \mathbf{0} \end{bmatrix},$$

which implies $\mathbf{W}^{\dagger}\boldsymbol{u} = \mathbf{0}$. However, $\mathbf{W}$ spans the center-subspace of $\mathbf{L}^{\dagger}$, and therefore $\mathbf{W}^{\dagger}\boldsymbol{u}$ is the projection of $\boldsymbol{u}$ onto the center-subspace of $\mathbf{L}$, which vanishes as required. □

A proof of corollary 1 in the case of infinite dimensional systems with parameter extensions can be found in [?].

II. CENTER MANIFOLD FOR THE EXTENDED SYSTEM

A. Normal Form

In the context of invariant manifold theory, most previous approaches for the approximation of such manifolds can be viewed effectively as instances of what is referred to as the parametrization method. The framework was introduced in a series of papers by [? ? ?], although, a particular instance of such an approach was presented earlier by [?] for the evaluation of the normal form of center manifolds. Nevertheless, subsequent works by [? ? ?] can be fruitfully viewed through the lens of the parameterization method. Within this framework the reduced representation of the invariant manifold being approximated can be chosen in an arbitrary manner depending on the particular requirements of the reduction being sought. Amongst the plethora of arbitrary choices, the normal form is an important representation that is often sought since it represents the simplest form of the reduced equations. In this context, it can be shown that for asymptotic normal form evaluation of the extended system the calculations involve only the resolvents of the original linear system and therefore all existing algorithms may be utilized for the calculations.

To simplify notation for multiple indices used hereafter, a multi-dimensional index set $\boldsymbol{\mathcal{I}}\{(i,j,k,\ldots),[a,b]\}$ is introduced, which is an ordered set of all multi-dimensional indices $(i,j,k,\ldots)$, defined as

$$\begin{aligned}\boldsymbol{\mathcal{I}}\{(i,j,k,\ldots),[a,b]\} :=& \boldsymbol{\{}(i,j,k,\ldots) \mid i,j,k,a,b \in \mathbb{N}, \\ & \forall\ a \leq i \leq b,\ i \leq j \leq b, \\ & j \leq k \leq b, \ldots \boldsymbol{\}},\end{aligned} \tag{33}$$

where, $i,j,k,\ldots$, are the index variables and $[a,b]$ specify the lower and upper bounds of the indices. Note that the higher index always starts from the previous index. Hence the index-set $\boldsymbol{\mathcal{I}}\{(i,j),[1,3]\}$ is the set containing the tuples, $\{(1,1),(1,2),(1,3),(2,2),(2,3),(3,3)\}$. Similarly, multiple nested summations are denoted as,

$$\sum_{(i,j,k,\ldots)}^{[a,b]} = \sum_{i=a}^{b}\sum_{j=i}^{b}\sum_{k=j}^{b}\cdots \tag{34}$$

As before, the higher index starts from the previous index.

Section I B and I C outline the spectral properties of the system when extended to include the parameters as dynamic variables. Effectively the size of the critical subspace increases and, the eigen-structure of the extended linearized system $\widehat{\mathbf{L}}$ can be deduced from the spectral properties of the original linear system $\mathbf{L}$. The critical subspace of the new system is of dimension $m = n+l+h$. The center-manifold theorem can be applied to this extended system given by equation (13) so that both parameter perturbations and harmonic forcings are directly incorporated into the larger center-manifold.

A particular graph style parameterization is chosen for the representation of the invariant center-manifold which explicitly decomposes the critical and stable subspace components. The solution to equation (13) is decomposed into the fields belonging to the critical and stable subspaces of $\widehat{\mathbf{L}}$. Since the critical subspace is finite dimensional, it is represented by the time-dependent amplitudes of the critical eigenvectors. Accordingly, the solution is decomposed as

$$\widehat{\boldsymbol{u}}(t) = \widehat{\mathbf{V}}\mathbf{x}(t) + \widehat{\mathbf{Q}}\widehat{\boldsymbol{u}}_s(t), \tag{35a}$$
$$\widehat{\mathbf{V}}\mathbf{x} = \widehat{\mathbf{V}}_{\mathbf{C}}\mathbf{x}_C + \widehat{\mathbf{V}}_{\mathbf{P}}\mathbf{x}_P + \widehat{\mathbf{V}}_{\mathbf{H}}\mathbf{x}_H, \tag{35b}$$

$$\mathbf{x} = \begin{bmatrix} \mathbf{x}_C \\ \mathbf{x}_P \\ \mathbf{x}_H \end{bmatrix}, \qquad \widehat{\boldsymbol{u}}_s(t) = \begin{bmatrix} \boldsymbol{u}_s \\ \mathbf{0} \\ \mathbf{0} \end{bmatrix},$$

where $\mathbf{x} \in \mathbb{C}^m$, is the complex vector of the time-dependent amplitudes of the critical eigenmodes, with $\mathbf{x}_C \in \mathbb{C}^n$, $\mathbf{x}_P \in \mathbb{C}^l$ and $\mathbf{x}_H \in \mathbb{C}^h$ being the amplitude vectors for the (extension of the) original critical modes, parameter modes and the harmonic modes respectively. $\widehat{\boldsymbol{u}}_s$ is the part of the solution lying in the stable subspace of $\widehat{\mathbf{L}}$. In accordance with lemma 1, the extended variables vanish for $\widehat{\boldsymbol{u}}_s$. $\widehat{\mathbf{Q}}$ is projector onto the stable subspace of $\widehat{\mathbf{L}}$. Note that $\widehat{\mathbf{Q}}\widehat{\boldsymbol{u}}_s = \widehat{\boldsymbol{u}}_s$, by definition of $\widehat{\mathbf{Q}}$ and $\widehat{\boldsymbol{u}}_s$. The nonlinear term is now denoted as $\mathcal{N}(\mathbf{x}, \boldsymbol{u}_s) = \boldsymbol{f}_{NL}(\boldsymbol{u}, \boldsymbol{\nu})$ with its extension $\widehat{\mathcal{N}}(\mathbf{x}, \boldsymbol{u}_s) = \widehat{\boldsymbol{f}}_{NL}(\boldsymbol{u}, \boldsymbol{\nu})$, to emphasize the change of variables. Substituting the solution decomposition in to equation (13) and using the projection matrices $\widehat{\mathbf{W}}$ and $\widehat{\mathbf{Q}}$ one obtains the two evolution equations for $\mathbf{x}$ and $\widehat{\boldsymbol{u}}_s$ as

$$\frac{d\mathbf{x}}{dt} = \widehat{\mathbf{K}}\mathbf{x} + \widehat{\mathbf{W}}^H \widehat{\mathcal{N}}. \tag{36a}$$

$$\frac{\partial \widehat{\mathbf{Q}}\widehat{\boldsymbol{u}}_s}{\partial t} = \widehat{\mathbf{Q}}\widehat{\mathbf{L}}\widehat{\mathbf{Q}}\widehat{\boldsymbol{u}}_s + \widehat{\mathbf{Q}}\widehat{\mathcal{N}}. \tag{36b}$$

Equation (36b) may be further simplified by noting that the individual terms have the following structure,

$$\widehat{\mathbf{Q}}\widehat{\boldsymbol{u}}_s = \begin{bmatrix} \mathbf{Q}\boldsymbol{u}_s \\ \mathbf{0} \\ \mathbf{0} \end{bmatrix}, \; \widehat{\mathbf{Q}}\widehat{\mathbf{L}}\widehat{\mathbf{Q}}\widehat{\boldsymbol{u}}_s = \begin{bmatrix} \mathbf{QLQ}\boldsymbol{u}_s \\ \mathbf{0} \\ \mathbf{0} \end{bmatrix}, \quad \widehat{\mathbf{Q}}\widehat{\mathcal{N}} = \begin{bmatrix} \mathbf{Q}\mathcal{N} \\ \mathbf{0} \\ \mathbf{0} \end{bmatrix},$$

Since the extended variables vanish for all three terms in equation (36b), the stable subspace solution $\boldsymbol{u}_s$ can be evaluated completely within the original system. A further notational change is made for the non-linear terms with,

$$\widehat{\mathcal{F}}(\mathbf{x}, \boldsymbol{u}_s) = \widehat{\mathbf{W}}^H \widehat{\mathcal{N}}(\mathbf{x}, \boldsymbol{u}_s),$$
$$\mathcal{G}(\mathbf{x}, \boldsymbol{u}_s) = \mathbf{Q}\mathcal{N}(\mathbf{x}, \boldsymbol{u}_s),$$

which results in the pair of equations given by

$$\frac{d\mathbf{x}}{dt} = \widehat{\mathbf{K}}\mathbf{x} + \widehat{\mathcal{F}}(\mathbf{x}, \boldsymbol{u}_s). \tag{37a}$$

$$\frac{\partial \mathbf{Q}\boldsymbol{u}_s}{\partial t} = (\mathbf{QLQ})\boldsymbol{u}_s + \mathcal{G}(\mathbf{x}, \boldsymbol{u}_s). \tag{37b}$$

The individual terms in equation (37) are easily interpretted. $\mathbf{x}$ is the vector if amplitudes of the critical eigenmodes of the extended space, $\widehat{\mathbf{K}}$ is the extended linearized operator reduced to the critical subspace and $\widehat{\mathcal{F}}$ is the vector of components of the projection of the nonlinearity in the critical eigendirections. $\boldsymbol{u}_s$ is the part of the solution that belongs to the stable subspace, $\mathbf{QLQ}$ is the restriction of the original linear operator to the stable subspace, and $\mathcal{G}(\mathbf{x}, \boldsymbol{u}_s)$ is the restriction of the nonlinearity to the stable subspace. The form of the equations is very similar to the one regularly used for center-manifold applications in ordinary differential equations [? ?].

Note that equation (37a) for the parameter and forcing modes simply reduces to,

$$\frac{d\mathbf{x}_P}{dt} = \boldsymbol{\Lambda}_P \mathbf{x}_P, \qquad \frac{d\mathbf{x}_H}{dt} = \boldsymbol{\Lambda}_H \mathbf{x}_H.$$

This represents equations (8b) and (8c) in the transformed variables. In addition, due to the chosen scaling of eigenvectors and use of cannonical vectors $\boldsymbol{e}_i^P$ for the definition of the parameter modes, the parameter perturbations are given by $\boldsymbol{\nu} = \mathbf{x}_P$. Similarly, for harmonic forcings in equation (7) we have $\boldsymbol{\theta} = \mathbf{x}_H$.

Finally note that the two evolution equations (37a) and (37b) represent slightly different quantities since, $\mathbf{x}$ is the critical mode *amplitude vector*. Equation (37a) is effectively the reduced order model of the system albeit, *not* in its normal form. $\boldsymbol{u}_s$ is the part of the total solution in the stable subspace, which is effectively driven by the critical subspace. Henceforth, $\boldsymbol{u}_s$ will generically be referred to as a field.

The center-manifold theorem is now applied to equation (37). The solution to equation (37), denoted as the pair $(\mathbf{x}, \boldsymbol{u}_s)$, is assumed to lie in a Hilbert space which is a direct product of the critical and stable subspaces, $\mathbb{H} = \mathbb{C}^m \bigoplus \mathbb{T}_s$, where $\mathbb{C}^m$ is the complex vector space of dimension m and represents the center subspace of the linearized operator of the system given by equation (37). $\mathbb{T}_s$ represents the stable subspace of the linearized operator. Using the center-manifold theorem, one may assume that the stable solution $\boldsymbol{u}_s$ evolves as a graph over the critical subspace $\mathbf{x}$ such that $\boldsymbol{u}_s \sim \mathfrak{h}(\mathbf{x})$ where, $\mathfrak{h}$ is a function with the following properties,

$$\mathfrak{h} : \mathbb{C}^m \to \mathbb{T}_s,$$
$$\mathfrak{h}(\mathbf{x}) \mapsto (\mathbf{0}, \mathbf{0}), \qquad \textit{for } \mathbf{x} = [0, 0, \ldots, 0],$$
$$\mathfrak{h}(\mathbf{x}) \sim \mathcal{O}(\mathbf{x}^2),$$

i.e., $\mathfrak{h}$ maps the critical subspace $\mathbb{C}^m$ into the stable subspace $\mathbb{T}_s$, vanishes at the origin and, is asymptotic to $\mathcal{O}(\mathbf{x}^2)$, as $\mathbf{x}$ approaches the origin. In general the smoothness of $\mathfrak{h}$ depends on the smoothness of the nonlinearities $\widehat{\mathcal{F}}$ and $\mathcal{G}$. Typically the center-manifold cannot be assumed to be analytic apriori [? ?]. In the current work degeneracies arising due to the loss of analyticity are not considered and the center manifold is assumed to be smooth enough to allow asymptotic approximations to a certain order. Substitution of $\mathfrak{h}$ in to equation (37) results in a graph equation,

$$\left(\frac{\partial \mathbf{Q}\mathfrak{h}}{\partial \mathbf{x}}\right) \cdot \left(\widehat{\mathbf{K}}\mathbf{x} + \widehat{\mathcal{F}}(\mathbf{x}, \mathfrak{h})\right) = (\mathbf{QLQ})\mathfrak{h} + \mathcal{G}(\mathbf{x}, \mathfrak{h}), \tag{38}$$

which defines the center-manifold for the system governed by equation (37).

B. Asymptotic Approximation

The graph equation for the solution in stable subspace obtained in equation (38) is in general a highly non-linear equation and typically can not be solved analytically. As an alternative, $\mathfrak{h}$ may be approximated asymptotically via a power series in $\mathbf{x}$ as,

$$\mathfrak{h}(\mathbf{x}) = \sum_{(a,b)}^{[1,m]} x_a x_b \boldsymbol{y}_{a,b} + \sum_{(a,b,c)}^{[1,m]} x_a x_b x_c \boldsymbol{y}_{a,b,c} + \mathcal{O}(\mathbf{x}^4), \tag{39}$$

where, the x_a, x_b, *etc*, are the (time-dependent) components of the critical subspace variables $\mathbf{x}$ and, the various $\boldsymbol{y}_{a,b,c...}$ are the *time-independent* fields associated with each term of the power series and represents the structure of the solution terms lying in the stable subspace. Substitution of the asymptotic expressions into the graph equation and with some algebraic manipulation, one obtains a series of linear inhomogeneous equations for the individual fields $\boldsymbol{y}_{a,b,c,...}$, which may be solved for, order by order. At each order k, one obtains $T_k = C(m+k-1,k)$ fields, where,

$$C(n,k) = \binom{n}{k} = \frac{n!}{k!(n-k)!}, \tag{40}$$

defines the binomial coefficient of n and k. For example for a critical subspace dimension of $m = 4$ one obtains $T_2 = 10$ unknown fields $\boldsymbol{y}_{a,b}$ at second order ($k = 2$) and, $T_3 = 20$ unknown fields $\boldsymbol{y}_{a,b,c}$ at third order ($k = 3$). Therefore one obtains a $T_k \times T_k$ matrix representing the linear system of equations for the fields $\boldsymbol{y}_{a,b,c,...}$, along with the inhomogeneous terms at each order k.

Considering a second-order asymptotic approximation for $\mathfrak{h}$, and the substitution of equation (39) in to equation (38), results in expressions for the second order terms as,

$$\begin{aligned}&\sum_{(a,b)}^{[1,m]}\sum_{i=1}^{m}(\widehat{K}_{a,i}x_i x_b + \widehat{K}_{b,i}x_a x_i)\mathbf{Q}\boldsymbol{y}_{a,b}\\ &-\sum_{(a,b)}^{[1,m]}[(\mathbf{QLQ})\boldsymbol{y}_{a,b} + \boldsymbol{g}_{a,b}]x_a x_b = 0,\end{aligned} \tag{41}$$

where, the terms arising from $\mathcal{G}$ and $[(\partial\mathbf{Q}\mathfrak{h}/\partial\mathbf{x})\cdot\widehat{\mathcal{F}}]$ are represented by $\boldsymbol{g}_{a,b}$. These do not contain any contributions from the field terms $\boldsymbol{y}_{a,b}$ and are the inhomogeneous terms arising at each order. $\widehat{K}_{a,i}$ and $\widehat{K}_{b,i}$ are the matrix elements of $\widehat{\mathbf{K}}$.

The various fields $\boldsymbol{y}_{a,b}$ are coupled due to the off-diagonal matrix elements of $\widehat{\mathbf{K}}$. This is in fact the case even at higher orders of the asymptotic approximation. It is always the off-diagonal terms of $\widehat{\mathbf{K}}$ that couple the different fields, $\boldsymbol{y}_{a,b,c,...}$, within each order, so that the system of equations obtained needs to be solved simultaneously for all fields within each order. For large systems, even at moderate values of m and k, this can become prohibitive since the number of fields T_k rises rapidly. However, the matrix $\widehat{\mathbf{K}}$ has a sparse and particular structure with $\boldsymbol{\Gamma}_P$ and $\boldsymbol{\Gamma}_H$ being responsible for the off-diagonal elements. Due to this structure the resulting fields may infact be solved sequentially. The structure of the system of equations arising at second order is analyzed symbolically in appendix **??**. Due to the sparsity pattern of $\widehat{\mathbf{K}}$, the system of equations obtained at second order results in a lower-triangular matrix for the unknown fields $\boldsymbol{y}_{a,b}$, which allows the fields to be solved individually in sequential order. This results in a general equation for the field terms at second order as,

$$(\widehat{\lambda}_\alpha + \widehat{\lambda}_\beta)\mathbf{Q}\boldsymbol{y}_{\alpha,\beta} - (\mathbf{QLQ})\boldsymbol{y}_{\alpha,\beta} = \boldsymbol{g}_{\alpha,\beta} - \boldsymbol{g}^c_{\alpha,\beta},$$

where, $\widehat{\lambda}_\alpha, \widehat{\lambda}_\beta \in \widehat{\boldsymbol{\lambda}}$, are the diagonal elements of $\widehat{\mathbf{K}}$, and the coupling terms for each pair of (α,β) have been denoted $\boldsymbol{g}^c_{\alpha,\beta}$. These coupling terms vanish identically for the indices $(\alpha,\beta) \in \boldsymbol{\mathcal{I}}\{(\alpha,\beta),[1,n]\}$) and for the remaining terms they are known apriori due to the lower-triangular nature of the system of equations. Recall that the unknown fields $\boldsymbol{y}_{\alpha,\beta} \in \mathbb{T}_s$, and $\mathbf{Q}$ is the projector into the stable subspace. Therefore $\mathbf{Q}^2 = \mathbf{Q}$, and, $\mathbf{Q}^2\boldsymbol{y}_{\alpha,\beta} = \mathbf{Q}\boldsymbol{y}_{\alpha,\beta}$ and the system may be represented in a slightly different manner as,

$$\begin{aligned}\boldsymbol{y}_{\alpha,\beta} =&\mathcal{R}_s(\widehat{\omega}_{\alpha,\beta})[\boldsymbol{g}_{\alpha,\beta} - \boldsymbol{g}^c_{\alpha,\beta}],\\ \widehat{\omega}_{\alpha,\beta} =&\widehat{\lambda}_\alpha + \widehat{\lambda}_\beta,\\ \mathcal{R}_s(\omega) =&[\mathbf{Q}(\omega\mathbf{I} - \mathbf{L})\mathbf{Q}]^{-1},\\ (\alpha,\beta) \in&\boldsymbol{\mathcal{I}}\{(\alpha,\beta),[1,m]\},\end{aligned} \tag{42}$$

where, the quantity $\mathcal{R}_s(\omega)$ is a *restricted resolvent* operator, *i.e.*, the resolvent operator restricted to the stable subspace of $\mathbf{L}$.

The lower-triangular structure of the system of equations for the unknown fields $\boldsymbol{y}_{\alpha,\beta,\gamma,...}$ persists for higher order terms as well. Therefore, if one considers a higher order power series for $\mathfrak{h}$, the structure of the solutions given by equation (42) can be generalized to higher orders, such that the expressions for the third and fourth order fields would be obtained as,

$$\begin{aligned}\boldsymbol{y}_{\alpha,\beta,\gamma,...} =&\mathcal{R}_s(\widehat{\omega}_{\alpha,\beta,\gamma,...})[\boldsymbol{g}_{\alpha,\beta,\gamma,...} - \boldsymbol{g}^c_{\alpha,\beta,\gamma,...}],\\ \widehat{\omega}_{\alpha,\beta,\gamma,...} =&\widehat{\lambda}_\alpha + \widehat{\lambda}_\beta + \widehat{\lambda}_\gamma + \ldots,\\ (\alpha,\beta,\gamma,\ldots) \in&\boldsymbol{\mathcal{I}}\{(\alpha,\beta,\gamma,\ldots),[1,m]\}.\end{aligned} \tag{43}$$

At all orders, the coupling terms $\boldsymbol{g}^c_{\alpha,\beta,\gamma,...}$ vanish for the index tuples $(\alpha,\beta,\gamma,\ldots) \in \boldsymbol{\mathcal{I}}\{(\alpha,\beta,\gamma,\ldots),[1,n]\}$. For higher order evaluations of the asymptotic series, the inhomogeneous terms $\boldsymbol{g}_{\alpha,\beta,\gamma,...}$ would also contain contributions from terms evaluated at lower order.

Finally, note that the resolvent operator in equations (42) and (43) is never singular. The singularity of the resolvent operator occurs at points on the complex plane corresponding to the eigenvalue spectrum of the corresponding operator. In this case the operator in question is the restricted operator ($\mathbf{QLQ}$), which has an eigenvalue spectrum with strictly negative real part ($\mathfrak{Re}(\lambda) < 0$). On the other hand, for all asymptotic terms, the restricted resolvent is evaluated at points $\widehat{\omega}$ which is the sum of integral multiples of the critical eigenvalues, hence $\mathfrak{Re}(\widehat{\omega}_{\alpha,\beta,...}) = 0$ identically. Therefore the restricted resolvent operator is non-singular for all the asymptotic terms. The singularity of the resolvent operator is usually associated with resonant terms leading to higher order terms in the Poincaré normal form or, to secular terms when considering a multiple time-scale expansion of solutions. In the current case the question of resonance never arises.

Finally, once the fields associated with the asymptotic expansion of $\mathfrak{h}(\mathbf{x})$ have been evaluated to the desired order, the power series for $\mathfrak{h}$ can be substituted into equation (37a) resulting in a set of ordinary differential equations for the evolution of the critical mode amplitudes in the center-manifold,

$$\frac{d\mathbf{x}}{dt} = \widehat{\mathbf{K}}\mathbf{x} + \widehat{\mathcal{F}}(\mathbf{x}, \mathfrak{h}(\mathbf{x})). \tag{44}$$

These are different from the usual "amplitude equations" [? ? ?] which typically refer to the slowly modulating component of solutions, also referred to as the Stuart-Landau equations [? ? ?] in the hydrodynamics literature. In this case the amplitude evolution includes both the fast time scale due to the instabilities as well as any slow timescale variations that may emerge. The Stuart-Landau equations may be derived from the reduced set of equations (37a) by considering a multiple-timescale expansion for the obtained ordinary differential equations, if so desired.